

Assignment-1: Implementation of a Multi-Client Banking Communication System Using Python Socket Programming.

A national bank has established a centralized transaction validation server that communicates with multiple ATM kiosks and branch terminals simultaneously. Each client sends transaction validation requests, such as account verification, balance enquiry, or transaction confirmation, to the server. The server must be capable of handling multiple client connections concurrently, process each client's request independently, and return the received message as an acknowledgment to confirm successful receipt. If one client disconnects, the server should continue providing uninterrupted service to the remaining connected clients.

Question:

Write a Python socket program to implement the above application using TCP sockets. Develop a multithreaded server capable of serving multiple clients simultaneously and a client program that connects to the server, sends multiple transaction validation messages, receives the echoed acknowledgment for each message, and terminates the connection when the user enters "exit".

Sample Input and Output

Banking Validation Server Started...

Server listening on 127.0.0.1:5000

Client Connected: ('127.0.0.1', 50120)

Client Connected: ('127.0.0.1', 50125)

[Client 50120] Received: Account Verification - ACC1001

[Client 50120] Echo Sent: Account Verification - ACC1001

[Client 50125] Received: Balance Enquiry - ACC2005

[Client 50125] Echo Sent: Balance Enquiry - ACC2005

[Client 50120] Received: Transaction Confirmation - TXN45879

[Client 50120] Echo Sent: Transaction Confirmation - TXN45879

[Client 50125] Received: Mini Statement Request

[Client 50125] Echo Sent: Mini Statement Request

Client ('127.0.0.1', 50120) disconnected.

[Client 50125] Received: Fund Transfer Validation

[Client 50125] Echo Sent: Fund Transfer Validation

Client ('127.0.0.1', 50125) disconnected.

Client 1 Output (ATM Terminal)

Connected to Banking Validation Server.

Enter Transaction Request: Account Verification - ACC1001

Server Acknowledgment:

Account Verification - ACC1001

Enter Transaction Request: Transaction Confirmation - TXN45879

Server Acknowledgment:

Transaction Confirmation - TXN45879

Enter Transaction Request: exit

Disconnected from Server.

Client 2 Output (Branch Terminal)

Connected to Banking Validation Server.

Enter Transaction Request: Balance Enquiry - ACC2005

Server Acknowledgment:

Balance Enquiry - ACC2005

Enter Transaction Request: Mini Statement Request

Server Acknowledgment:

Mini Statement Request

Enter Transaction Request: Fund Transfer Validation

Server Acknowledgment:

Fund Transfer Validation

Enter Transaction Request: exit

Disconnected from Server.

*****All the Best*****